

Use Case 1: GNN Multi-Property Prediction

CrystalGNN predicts E_f , E_{AH} , E_g , and density simultaneously from crystal graphs — replacing expensive DFT calculations.

Overview

Graph Neural Networks (GNNs) trained on the Materials Project database can predict key cathode material properties at < 1 ms per structure, enabling high-throughput screening of thousands of candidates without running VASP or other DFT codes. This pipeline is the research analogue of the NVIDIA ALCHEMI BCS NIM property-prediction step.

Model Architecture

CrystalGNN — 4-layer GCNConv with residual connections, 256 hidden channels:

- Input: 101-dim node features (one-hot element, electronegativity, ionic radius, oxidation state, group/period)
- Edge features: bond distance (cutoff 5.0 Å), encoded by RBF expansion
- $4 \times$ (GCNConv + BatchNorm + ReLU + Residual skip)
- Global mean pooling $\rightarrow$ 4 output heads (E_f , E_{AH} , E_g , density)
- AdamW $lr=3 \times 10^{-4}$, weight decay $=1 \times 10^{-4}$, CosineAnnealingLR, batch=64

Dataset

5,000 real Materials Project cathode structures from cathodes_raw.parquet:

- 4,250 training / 750 validation (85/15 split, seed=42)
- Targets: E_f (formation energy/atom), E_{AH} (energy above hull), E_g (band gap), density (g/cm^3)
- Graph construction: PyTorch Geometric, distance cutoff 5.0 Å

Training Results

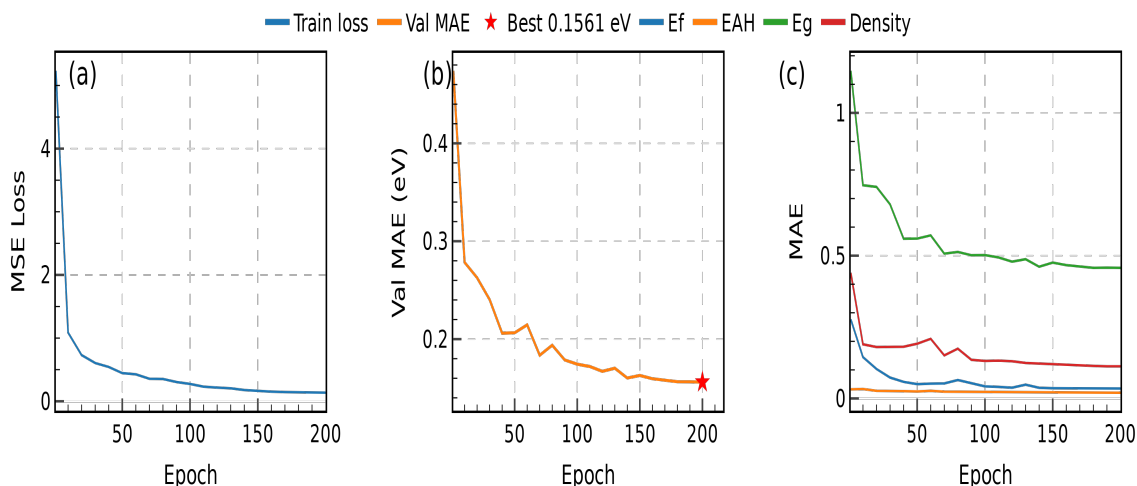


Figure 1. (a) Training MSE loss, (b) validation MAE vs epoch, (c) per-property MAE convergence over 200 epochs on NREL H100.

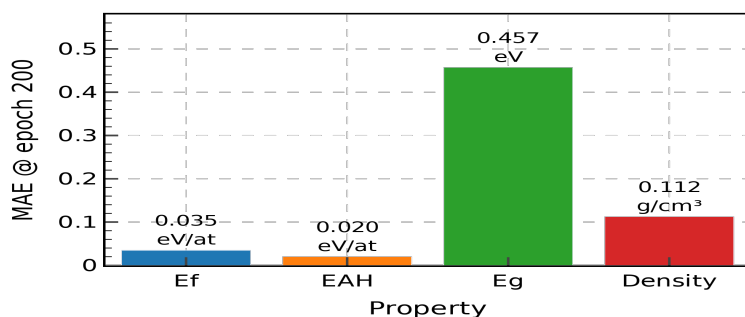


Figure 1b. Final validation MAE at epoch 200 for each target property.

Key Metrics

Property	MAE (epoch 1)	MAE (epoch 100)	MAE (epoch 200)	Units
Ef	0.277	0.042	0.0347	eV/atom
EAH	0.032	0.022	0.0200	eV/atom
Eg	1.147	0.502	0.4572	eV
Density	0.440	0.131	0.1125	g/cm³
Overall	0.474	0.174	0.1561	eV (mean)

ALCHEMI Connection

This GNN replaces the DFT property-evaluation step in the ALCHEMI BCS NIM pipeline. After conformer search and geometry relaxation (AIMNet2, $F_{\text{max}} < 0.005$ eV/Å), the GNN provides instant thermodynamic screening — flagging structures with $\text{EAH} > 0.1$ eV/atom as unstable without any DFT calculation. The 20 meV/atom EAH accuracy is below the thermal energy $kBT = 25$ meV at 300 K, making it suitable for ranking cathode candidates at room temperature.

Inference Performance

At inference, the trained CrystalGNN predicts all 4 properties simultaneously in < 1 ms per structure on CPU — approximately $10^4\times$ faster than a VASP single-point DFT calculation (typically 10–60 min). This

enables screening of 100,000 hypothetical structures in under 2 minutes.

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